Practice runs and evidence

antocciphysics | TEACHER KEY | CPA

Learning target: Use repeated trials to prepare an evidence-based design explanation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 5 course days later.*
Ideal speed becomes 5 times as large. Actual rpm, load or slipping may change.

02. Explain a worked example (worked)
All three retained; mean≈2.17 s, fastest 2.0 s.

03. Look and explain (model)
Actual evidence with functional labels and explanations.

04. Build the solution (practice)
.80 m/s for supplied example; actual values must be measured.

05. Use the idea (reason)
Retain and annotate failure; report completed-run speeds separately with inclusion criteria and failure count.

06. Choose a method (mixed) *Review: Unit 7, Day 14; 12 course days later.*
Faster flux change or more turns increases emf magnitude. Stronger field/effective area changes are valid with suitable controls.

07. Exit check (exit)
Student cites actual evidence and acknowledges if evidence is limited.

08. STRETCH: Apply it another way (stretch)
Second set faster mean 2.1 s and more consistent; first has fastest individual run 2.0 s. Choice depends on stated criteria.

Teaching note: Final approval, signature and photo follow classroom requirements. Do not invent trial values to fill blank rows.

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01. Bring back an earlier idea (retrieve) *Review: Unit 8, Day 3; 5 course days later.*
Torque ratio 7; speed ratio 1/7. Assumes ideal gears with negligible losses.

02. Explain a worked example (worked)
All three retained; mean≈2.17 s, fastest 2.0 s.

03. Represent and interpret (model)
Actual evidence with functional labels and explanations.

04. Solve and justify (practice)
Mean 5.0 s; range .4 s; .80 m/s from mean time.

05. Analyze the reasoning (reason)
Retain and annotate failure; report completed-run speeds separately with inclusion criteria and failure count.

06. Choose a method (mixed) *Review: Unit 7, Day 14; 12 course days later.*
Emf is 4 times as large. Same turn count and flux change per turn; compare average emf over each interval.

07. Exit check (exit)
Claim is bounded; numerical/observational evidence named; physical mechanism fits; causal certainty matches the comparison.

08. STRETCH: Apply it another way (stretch)
Mean speeds (1.5+1.0)/2=1.25 m/s; total 6/5=1.2 m/s. Equal-trial and time-weighted averaging differ.

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